■ Scratch — Mini Catch Game

Let's learn coding by making a fun catching game!

Grade 2 • Beginner • Scratch 3.0

■ **Subject**	Computer Science / Coding
■ **Goal**	Build a fruit-catching game in Scratch
■ **Duration**	45–60 minutes
■ **Tool**	Scratch 3.0 (scratch.mit.edu) — free, no login needed
■ **Sprites**	Bowl (catcher) + Fruit (falling object)

■ Learning Outcomes

- Understand what coding is and why we use it
- Navigate the Scratch interface (Stage, Sprites, Code Area)
- Use Motion, Events, Control, Sensing, and Variable blocks
- Build a complete working game step-by-step
- Test and debug their own code

What is Coding?

Coding is telling the computer what to do by giving it **step-by-step instructions**.

Just like you follow steps to make a sandwich, computers follow steps *we* write!

Coding helps us create **games**, **stories**, and **solve problems**.

Why Scratch?

Scratch uses **colourful blocks** to code. No typing needed — just **drag and drop**!

■ **Create fun games**	Build interactive games with points & sound
■ **Make animations**	Bring sprites to life with movement
■ **Tell stories**	Create interactive storybooks
■ **It's FREE**	scratch.mit.edu — no login required for students

Meet the Scratch Interface

The Scratch screen has **3 main parts**:

1■■ Stage	Where your game plays — the big screen in the top-left
2■■ Sprite List	Your characters appear here below the Stage
3■■ Code Area	Drag coloured blocks here to write your code

■ **Tip:** Click the **green flag** ■ to run your code. Click the **red stop** ■ to end it.

Sprites — Your Game Characters

Sprites are **characters or objects** in your game. For our catch game we need:

■ **Bowl sprite**	The catcher — moves left and right at the bottom
■ **Fruit sprite**	The falling object — drops from the top of the screen

You can pick sprites from the **Scratch library** or draw your own! Click the cat icon at the bottom-right to choose a sprite.

Coding Blocks We Will Use

Scratch has colourful blocks for coding. We drag and snap blocks together like **puzzle pieces**!

BLOCK	**Motion (blue)**	Move sprites — change x, change y, go to position
BLOCK	**Events (yellow)**	Start actions — When green flag clicked, When key pressed
BLOCK	**Control (orange)**	Repeat and check — forever loop, if … then
BLOCK	**Sensing (cyan)**	Detect touching — if touching bowl?
BLOCK	**Variables (red)**	Keep score — set Score to 0, change Score by 1

What is the Mini Catch Game?

■ **Objects fall**	Fruit drops from the top of the screen
■ **You move**	Use arrow keys to slide the bowl left and right
■ **You score**	Catch the falling fruit to earn points!
■ **Keep going**	The more you catch, the higher your score

Let's build this game **step by step** using Scratch blocks!

Building the Game — Step by Step

Step 1: Start the Game

Every Scratch game starts with the **When green flag clicked** block.

This block tells the computer: *"Begin the game now!"*

Blocks to use:

```
When green flag clicked
Set [Score] to (0)
```

We set the score to **0** so the game starts fresh every time.

Step 2: Move the Bowl

We use **arrow keys** to slide the bowl left and right.

Blocks to use:

```
When [right arrow] key pressed → Change x by (10)
When [left arrow] key pressed → Change x by (-10)
```

Changing **x** moves the sprite horizontally. Positive = right, negative = left.

Step 3: Make the Fruit Fall

The fruit starts at the top and falls down. We use a **forever loop** to keep it falling!

Blocks to use:

```
When green flag clicked
Go to x: (pick random -200 to 200) y: (170)
Forever
Change y by (-5)
If (y position) < -170 then
Go to x: (pick random -200 to 200) y: (170)
```

If the fruit reaches the bottom (y < -170), it jumps back to the top.

Step 4: Catch the Fruit!

The game detects a catch using the **sensing block**.

Blocks to use (inside the forever loop):

```
If then
Change [Score] by (1)
Go to x: (pick random -200 to 200) y: (170)
```

When the fruit touches the bowl, the score goes up by 1 and the fruit resets to the top!

Step 5: Keep Score

We need a **variable** to count points.

How to create the Score variable:

1. Click **Variables** in the block menu
2. Click **Make a Variable**
3. Name it **Score** and click OK

Every time you catch a fruit:

```
Change [Score] by (1)
```

The score is displayed on the Stage so players can see their points!

Step 6: Add Fun Sounds!

Make your game exciting with **sounds** from the Scratch library!

Block to use (inside the 'if touching' block):

```
Play sound [Pop] until done
```

You can also try **Chomp** or **Collect** for different effects. Go to the Sounds tab to pick from the library.

Test Your Game!

Time to play! Click the **green flag** and check these:

■	Does the bowl move left and right with the arrow keys?
■	Does the fruit fall from the top of the screen?
■	Does the score go up by 1 when you catch the fruit?
■	Does the fruit reset to the top after being caught?
■	Does a sound play when you catch?

■ If all boxes pass — your game works! You are a coder now!

Fix Common Problems

■ **Bowl not moving?**	Check your arrow key event blocks. Make sure the bowl sprite is selected in the Sprite List.
■ **Fruit not falling?**	Check the forever loop and the 'Change y by -5' block. Make sure the fruit sprite is selected.
■ **Score not counting?**	Check the 'If touching [Bowl]?' block. Make sure you named the variable **Score**.
■ **No sound playing?**	Go to the Sounds tab and add a sound from the library first, then use the Play sound block.

Ask your teacher if you need help! ■

What We Learned Today

■ **Sprites**	Characters or objects in a game
■ **Motion blocks**	Move sprites left, right, up, or down
■ **Loops**	Make things repeat (forever loop)
■ **Sensing blocks**	Detect when two sprites touch
■ **Variables**	Keep track of numbers like the score

■ **Sounds**	Add audio reactions to events

■ Great job! You built a real game in Scratch. You are a coder now! Keep exploring and make it even better. ■